Renewable Energy 99 (2016) 45–56

Contents lists available at ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

The effect of channel constriction on the potential for tidal stream power

Malcolm Smeaton*, Ross Vennell, Alice Harang

Ocean Physics Group, Department of Marine Science, University of Otago, Dunedin, New Zealand

ARTICLE INFO

Article history:
Received 7 August 2015
Received in revised form
23 April 2016
Accepted 6 June 2016
Available online 27 June 2016

Keywords:
Tidal power
Tidal energy
Channel constriction
Channel potential
Channel geometry

ABSTRACT

Tidal channels with narrow constrictions are attractive sites for electricity generation due to energy dense, high velocity flow through the constricted zones. This work uses a 1D model to systematically examine the effect of constriction on channel potential (theoretical maximum power) and transport in channels connecting two large water bodies (Type I) and channels connecting a large water body to an embayment/lagoon (Type II). Type I channels showed a monotonic decrease in potential and transport with increasing constriction due to enhanced bottom drag from the resultant high velocity zone. The effect of constriction on potential and transport in Type II channels varies depending on the relative geometries of the channel and embayment. Type II channels may be geometrically modified to increase power and in some instances this would simultaneously produce a high velocity zone. The flow reduction required to achieve a channel's potential was invariant to changes in constriction for Type I channels and also Type II channels if a "lagoon geometry factor" is less than about 4. A simple approximation for potential provided in the literature for unconstricted channels is extended for use with constricted channels and an approximation for the drag coefficient required to achieve potential is provided.

© 2016 Elsevier Ltd. All rights reserved.

1. Introduction

Electricity generation from strong flows through tidal channels could potentially make a significant contribution to the world's renewable energy supply [4,8,15,17,18]. Constricted channels are potentially lucrative sites for energy development due to enhanced tidal velocities occurring where flow is forced through the narrowest section. It may be possible, in these high velocity zones, to generate the same amount of power using fewer turbines than would be required outside of the channel's constriction. This may also be economically, socially and environmentally beneficial in some cases. Little electricity is generated from tidal currents at present due to high initial investment costs and a reasonable degree of uncertainty surrounding the future of the industry. Enhanced understanding of the tidal resource and improved resource assessment methods can help reduce some of this uncertainty and increase investor confidence.

This work extends our understanding of the tidal resource beyond the case of an unconstricted, rectangular channel. The works of [2,6,8] made several conclusions about the potential and flow rate within unconstricted, rectangular channels but did not explore the effect of channel constriction or variable cross-section. Furthermore, the studies of [2,6,8] did not describe the relationship between channel potential and geometry nor identify with any certainty how a constriction may be used to advantage.

This work makes what may be the first systematic look at how tidal power is affected by channel constriction. Channel potential is the theoretical upper limit to the amount of power that can be lost by a channel to a tidal turbine array and can be realised by occupying the channel's cross-section with a wall of turbines [7]. While this is likely to be an unattainable goal in most parts of the world due to environmental, economic and social reasons; channel potential provides a useful first indication of the amount of available power at a given site.

Multi-dimensional modelling of different geometries has been completed for site-specific resource assessment studies in the past [3–5,11] however little work has been completed on the general relationship between channel geometry and tidal power. While complex numerical models can provide a great deal of information to a reasonably high degree of accuracy, they are usually

* Corresponding author.
E-mail addresses: smema798@student.otago.ac.nz (M. Smeaton), ross.vennell@otago.ac.nz (R. Vennell), alice.harang@otago.ac.nz (A. Harang).

http://dx.doi.org/10.1016/j.renene.2016.06.013
0960-1481/© 2016 Elsevier Ltd. All rights reserved.

computationally demanding and gathering results is time consuming. One-dimensional (1D) modelling allows for results to be collected quickly for a myriad of conditions. Although less realistic and detailed, one-dimensional modelling is particularly suitable for systematic studies such as this one. This work uses the models derived in Refs. [2,6,8] along with an analytical solution provided by Ref. [16] to examine the effect of channel constriction on channel potential and channel dynamics.

A derivation and outline of the model used for this study is given below in Section 2. The effects of channel constriction on channel potential are then presented in Section 4 for both channel types. Section 5 then proceeds to look at how the degree of channel constriction influences flow through the channel and how the flow reduction required to achieve potential varies with channel geometry. Finally, a simple approximation provided in Ref. [6] for unconstricted Type I channels - extended by Ref. [2] to apply to unconstricted Type II channels - is then further extended here to apply to constricted cases. A similar approximation is also presented that allows the farm drag required at potential to be quickly approximated for shallow, friction dominated channels.

2. Tidal model

2.1. Dynamic model for open ended channels (type I)

Consider a tidal channel connecting two large water bodies of variable rectangular cross-section $A(x)$. The x axis spans the length of the channel beginning at the first water body ($x = 0$) and terminating at the second water body ($x = L$). Tidal currents are driven by the difference in free surface elevation between the two water bodies. Two sources of friction exist in the channel; the first pertains to bottom friction associated with the sea floor and the second is that associated with the addition of turbines. Starting with the dynamical governing equation as given in Ref. [6] and following steps taken in Ref. [16]:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = -g\frac{\partial \eta}{\partial x} - \frac{C_D}{h}|u|u - F_T. \tag{1}$$

Where $u(x,t)$ is the velocity of the flow which is assumed to be uniform at any given cross-section of the rectangular channel, g is the acceleration due to gravity, $\eta(x,t)$ is the displacement of the water's free surface from depth $h(x)$, C_D is the background bottom drag coefficient and F_T is an additional force associated with the presence of a turbine farm. The momentum balance for the channel as a whole can be found by multiplying by $\rho A(x)$ and integrating across the length of the channel.

$$\rho\int_0^L A\frac{\partial u}{\partial t}\mathrm{d}x + \rho\int_0^L Au\frac{\partial u}{\partial x}\mathrm{d}x = -\rho g\int_0^L A\frac{\partial \eta}{\partial x}\mathrm{d}x - \rho C_D\int_0^L \frac{A}{h}|u|u\mathrm{d}x - \rho\int_0^L AF_T\mathrm{d}x. \tag{2}$$

If the channel is dynamically short then the volume flow rate of the channel is approximately constant across its length and varies only with time e.g. $Q(t) = u(x,t)A(x)$ [12,13]. Expressing the velocity in (1) with respect to flow rate and cross-sectional area then gives:

$$\left(\int_0^L \frac{\mathrm{d}x}{A}\right)\frac{\mathrm{d}Q}{\mathrm{d}t} + \frac{Q^2}{2}\left[\frac{1}{A_L^2} - \frac{1}{A_0^2}\right] = g[\eta_0 - \eta_L] - C_D\left(\int_0^L \frac{\mathrm{d}x}{hA^2}\right)|Q|Q - \int_0^L F_T\mathrm{d}x. \tag{3}$$

The turbine farm is assumed to obey a quadratic drag law, thus the integral of F_T may be defined as:

$$\int_0^L F_T\mathrm{d}x = \frac{C_F}{A_F^2}|Q|Q. \tag{4}$$

Where A_F is the cross sectional area at the location of the farm and C_F is the gross farm drag coefficient. Substituting this into (3) gives:

$$\left(\int_0^L \frac{\mathrm{d}x}{A}\right)\frac{\mathrm{d}Q}{\mathrm{d}t} + \frac{Q^2}{2}\left[\frac{1}{A_L^2} - \frac{1}{A_0^2}\right] = g[\eta_0 - \eta_L] - \left(C_D\left(\int_0^L \frac{dx}{hA^2}\right) + \frac{C_F}{A_F^2}\right)|Q|Q. \tag{5}$$

For simplicity, it is assumed that the cross-sectional areas at both ends of the channel are the same and thus the advection term in (5) vanishes. The two remaining integrals in (5) are referred to in the literature as the channel's geometric factors; $i = \int_0^L \frac{dx}{A}$ and $b = \int_0^L \frac{dx}{hA^2}$. Expressing the difference in free surface between the two terminal water bodies as a sinusoidal function of time (t) with angular frequency (ω) and amplitude (Δ), Equation (5) becomes:

$$i\frac{dQ}{dt} = g\Delta\sin(\omega t) - \left(bC_D + \frac{C_F}{A_F^2}\right)|Q|Q. \tag{6}$$

In the absence of friction, for an unconstricted channel of uniform rectangular cross-section, Equation (6) has an exact solution $Q(t) = Q_{IU}\cos(\omega t)$ where $Q_{IU} = \frac{g\Delta A_0}{\omega L}$. Flow rate can be non-dimensionalised using this value and time using $t' = t\omega$. Depth and area can then be non-dimensionalised with respect to h_0 and A_0 (their maximum values that occur outside of the constricted zone). Using the non-dimensional parameter $\widehat{\alpha} = 8g\Delta/3\pi\omega^2L^2$ [16] then allows the non-dimensional bottom drag and farm drag to be defined as $\lambda_0 = \widehat{\alpha}LC_D/h_0$ and $\lambda_F = \widehat{\alpha}C_F/A_F'^2$ respectively. We then define the effective bottom drag as $B\lambda_0$ which when added to the farm drag constitutes the total drag coefficient of the channel: $\lambda_T = B\lambda_0 + \lambda_F$. Equation (6) in non-dimensional form is thus:

$$I\frac{\mathrm{d}Q'}{\mathrm{d}t'} = \sin(t') - \frac{3\pi}{8}\lambda_T|Q'|Q'. \tag{7}$$

The $\frac{3\pi}{8}$ factor is introduced for mathematical convenience at a later point and $I = \int_0^1 \frac{dx'}{A'}$ and $B = \int_0^1 \frac{dx'}{h'A'^2}$ are the non-dimensional equivalents of the geometric factors introduced in (6). Note that although Equation (7) is referred to as a "1D model", strictly speaking it is a "0D model" as it has been derived by integrating along the length of the channel.

2.2. Channels connected to a lagoon or embayment (type II)

For channels connecting a large water body to a bay of finite volume as described in Refs. [2,8], the tidal head difference driving

transport through the channel is now dependent on the water level inside the bay which is dependent on the flow rate itself. Following the same derivation as described in Section 2.1, Equation (8) is the lagoon channel equivalent to (6). If the embayment is assumed to be prismatic then the free surface elevation inside the bay is related to flow rate and it's surface area A_s by (9)

$$i\frac{dQ}{dt} = g\left(\eta_{ocean} - \eta_{bay}\right) - \left(bC_D + \frac{C_F}{A_F}\right)|Q|Q, \tag{8}$$

$$\frac{d\eta_{bay}}{dt} = \frac{Q}{A_s}. \tag{9}$$

Here, $\eta_{ocean} = \eta_0 \sin(\omega t)$ is the sinusoidally varying free surface elevation of the large water body and η_{bay} is the time varying free surface of the bay. Introducing the parameter $\widehat{\alpha}^* = 8g\eta_0/3\pi\omega^2L^2$ allows for the non-dimensional bottom drag and farm drag for lagoon channels to be expressed as $\lambda_0^* = \widehat{\alpha}^* LC_D/h_0$ and $\lambda_F = \widehat{\alpha}^* A_0 C_F/A_F'$ respectively. The total drag in the channel is then $\lambda_T^* = B\lambda_0^* + \lambda_F^*$. Non-dimensionalising against an unconstricted, frictionless Type I channel, Equations (8) and (9) are respectively:

$$I\frac{\mathrm{d}Q'}{\mathrm{d}t'} = \sin(t') - \eta_{bay}' - \frac{3\pi}{8}\lambda_T^*|Q'|Q', \tag{10}$$

$$\frac{\mathrm{d}\eta_{bay}'}{\mathrm{d}t'} = \beta Q'. \tag{11}$$

Where $\beta = (g/\omega^2)(A_0/LA_s)$ is referred to as the lagoon factor and is the ratio of the Helmholtz frequency of the basin to the frequency of the forcing tides. From here onwards the * is dropped from the dimensionless drag coefficients for lagoon channels and it is assumed that the subtle differences in drag coefficient definitions of the two channel types are understood.

2.3. Channel constrictions

Constrictions used in this study are Gaussian shaped. A width constriction is made of two symmetrical Gaussian curves with the line of symmetry spanning the length of the channel at $W/2$. Depth constrictions are based on a single Gaussian sill. Both width and depth constrictions are formulated so that their respective curves peak halfway along the channel's length at $x = L/2$. The curve used to constrict the channel width is:

$$G_w(x') = \frac{W_c}{2}\exp\left[\frac{-\left(x' - \frac{1}{2}\right)^2}{c^2}\right]. \tag{12}$$

Where W_c is the width constriction factor and is the width of the "land mass" at the most constricted point of the channel relative to the flow width outside of the constriction (W_0). Thus, for a constriction that at most blocks the channel by 60%, $W_c = 0.6$. The $\frac{1}{2}$ in the numerator of the exponential function ensures the curve peaks at the midpoint of the channel length. Finally, the factor c controls the broadness of the constriction such that as $c \to \infty$, $G_w(x') \to \frac{W_c}{2}$. In this study, c is set at 0.1. As the width constriction consists of two of these curves, the non-dimensionalised width of the channel flow area at any point is $W'(x') = 1 - 2G_w(x')$.

Depth constrictions are based on a similar curve:

$$G_h(x) = h_c \exp\left[\frac{-\left(x' - \frac{1}{2}\right)^2}{c^2}\right]. \tag{13}$$

Here h_c is the depth constriction factor and is analogous to the width constriction factor but in the depth axis. The remaining variables in (13) are the same as in (12). As the depth constriction consists of only one of these curves (Fig. 2), the total depth at any point in the channel is given by $h'(x') = 1 - G_h(x')$. Again, we use $c = 0.1$. Note that a width constriction factor and depth constriction factor of equal value will result in the same flow area reduction along the channel length and decrease the overall flow volume of the channel by the same amount. Furthermore, any specific constriction factor will result in the same flow area and flow volume reductions regardless of whether the channel is Type I or Type II.

2.3.1. Flow separation

As we do not consider the effects of flow separation, our model is only applicable to gradual constrictions. Flow separation may occur at the lee side of the constriction where the expanding cross-section will result in an adverse pressure gradient which will cause flow in the boundary layer to decelerate to the point of reversal. The presence of bottom drag will further enhance this effect. Depending on the aspect ratio of the constriction geometry, the frictional Reynolds number and the Keulegan-Carpenter number âĂŞ this separated flow may cause eddies to form [10]. Both flow separation alone and flow separation leading to eddy formation will result in a loss of energy in the channel. Quantifying this energy loss is beyond the scope of this study and needs more investigation with a 2D or 3D model. We plan to do this in a subsequent work.

2.3.2. Channel geometric factors (I and B) and multiple constrictions

The channel geometric factors both increase from an initial value of 1 where the channel is unconstricted to a value of infinity when the channel is fully blocked off (Fig. 3). The geometric factor I is the same for both constriction types (e.g. depth or width). The factor B is more sensitive to constriction factor than I due to its dependence on the square of the area and is also sensitive to depth constriction due to its dependence on h. This study has limited itself to looking at channels with a single Gaussian shaped constriction as described in the preceding section. The effect of having multiple constrictions within this one-dimensional model would cause the I and B factors to further increase. Thus, with respect to the channel potential and transport, the effect of several small constrictions would be identical to that of a single large constriction that produced the same values of I and B.

2.3.3. Constriction and Froude number

This model is only valid for small Froude numbers. In some extreme constriction cases the Froude number ($Fr = u/\sqrt{gh}$) might approach a value of 1, resulting in critical flow in the channel. As the flow accelerates through the constricted region, Fr will increase via an increase in the flow velocity (u). This increase will be more profound if the constriction is a depth constriction as not only will the velocity of the flow increase, but the depth (h) of the channel will decrease. As is discussed later on in this work (Section 5), provided $I > \beta$, increasing the degree of constriction in a channel will cause the transport to decrease. This effect may partially offset the increase in Fr.

3. Analytical approximation

[16] provides an approximate analytical solution to (7) that agrees closely with solutions obtained through numerical methods [2,6,8]. The analytical solution considers only the M_2 constituent and assumes the transport can be expressed as a single frequency sinusoidal current (14):

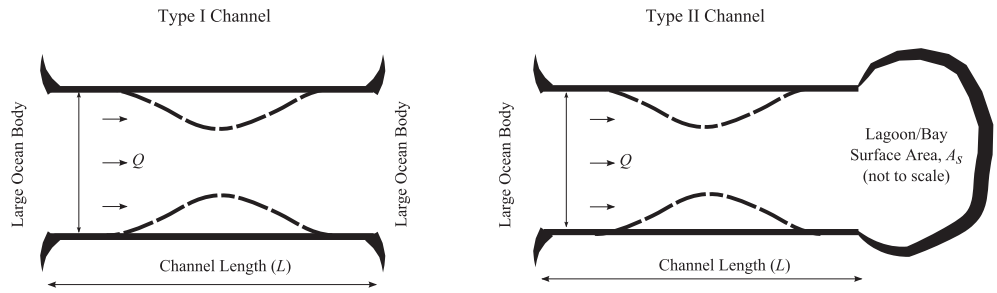

Fig. 1. Illustrations showing a Type I channel connecting two infinitely large water bodies (left) and a Type II channel connecting one infinitely large water body to a prismatic lagoon of surface area A_s (right). The dashed line shows how a width constriction would be built into the channels using our model. Q is the time varying transport through the channel and L is the channel's length. Note that in reality the size of the lagoon/bay is much greater than the size of the channel.

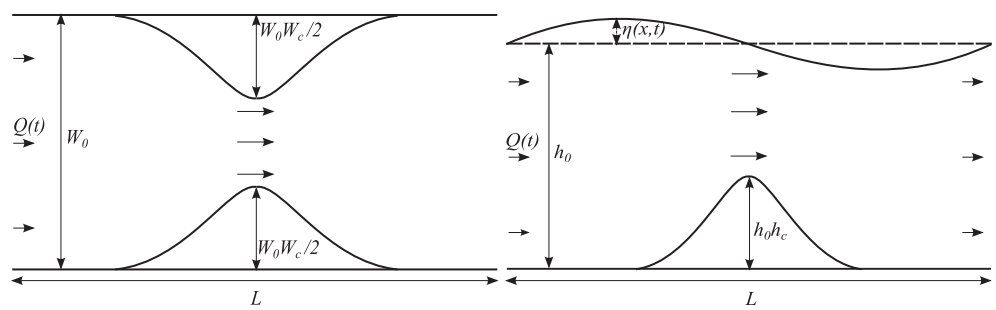

Fig. 2. Examples of width (left) and depth (b) constriction formulations used in this study. Width constrictions are based on two symmetric Gaussian curves (Equation (12)) while depth constrictions consist of a single Gaussian sill (Equation (13)). Constriction factors (W_c and h_c) determine the extent to which the constriction occupies the cross-section at the channel's narrowest point.

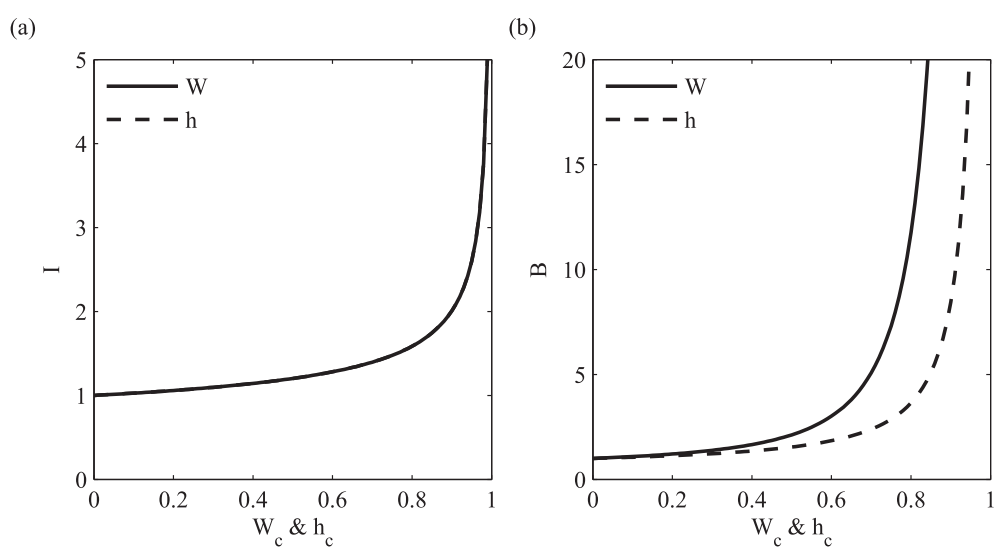

Fig. 3. Effect of Depth (h_c) and Width (W_c) Constrictions on the channel's non-dimensional geometric factors $I = \int_0^1 \frac{dx'}{A'} O$ and $B = \int_0^1 \frac{dx'}{h'A'^2}$. Constriction factor is defined as the ratio of land area at the most constricted point to the flow area outside of the constriction e.g. a channel with a constriction factor of 0.8 will have 80% of its cross sectional area occupied by constriction at the most constricted point.

$$Q' = Q_0 \sin(t' - \varphi_u). \tag{14}$$

Where φ_u is the phase difference between the channel transport and the channel forcing. This forcing is driven by differences in phase and amplitude of the water surface at the ends of the channel. The non-linearity of the bottom friction term introduces higher harmonics which distort the transport function from a perfect sine wave. This distortion becomes important when analysing power over a full spring/neap cycle and this effect is explored in detail by Ref. [1]. The error in using this analytical approximation instead of solving the problem numerically is less than 4% [16].

Substituting (14) into (7) and using the approximation $|Q|Q \approx \frac{8}{3\pi} Q_0 \sin(t' - \varphi_u)$ [15] yields two simultaneous equations of which the first can be solved to give an expression for φ_u. This can subsequently be used with the second equation to obtain an expression for Q_0. Similarly, substituting (14) into (10) gives an

analytical solution for Type II channels (Equations (15) and (16)). The analytical approximation for Type II channels can be used for Type I channels by setting $\beta = 0$. Equations (15) and (16) are the generalised forms of those given in Ref. [2] who considered only an unconstricted, rectangular channel where $I = 1$:

$$\varphi_u = \tan^{-1}\left(\frac{I-\beta}{\lambda_T Q_0}\right), \tag{15}$$

$$Q_0 = \frac{\sqrt{2}}{\sqrt{\sqrt{4\lambda_T^2 + (I-\beta)^4} + (I-\beta)^2}}. \tag{16}$$

4. Effects of channel constriction on channel potential

This study uses channel potential as a proxy to examine the effects of channel constriction on tidal stream power production. Channel potential is the theoretical upper limit for power generation in a tidal channel and can be achieved with a single row of turbines occupying 100% of the channel's cross-section. Channel potential considers the power in the channel that can be lost to all sources of farm related drag and therefore includes power production, mixing losses behind the turbines, structural drag related losses and electromechanical losses [16]. The amount of channel potential that can actually be realised is dependent on turbine array design and configuration [18]. The number of rows of turbines, the number of turbines in each row, how the turbines are laid out in these rows and the blade pitch of the turbines have all been shown to have significant influence on the amount of power lost to a turbine array [14,16,18]. While channel potential does give an overestimate of the power that can be generated by an array, it is a useful quantity for the purpose of observing channel scale responses to changes in channel geometry. The non-dimensionalised potential ($\overline{P}'^{opt}_{farm}$) is calculated by finding the optimal non-dimensional farm drag (λ_F^{opt}) which maximises the average power lost to the farm over a tidal cycle for some geometric factors I and B (17).

$$\overline{P}'^{opt}_{farm} = \frac{1}{2\pi}\int_0^{2\pi} \lambda_F^{opt}|Q'(t')|^3 \mathrm{d}t'. \tag{17}$$

Substituting the analytical solution provided in (16), into (17) gives:

$$\overline{P}'^{opt}_{farm} = 2\left(\lambda_T^{opt} - B\lambda_0\right)Q_0^3. \tag{18}$$

4.1. Increasing constriction reduces the potential of type I channels

This study was done entirely in non-dimensional variables for a hypothetical channel. Although this study did not model any specific channels we supply Table 1 to give some perspective to the figures we have used throughout this work.

Fig. 4(a) shows the response of channel potential to increases in constriction for three different values of bottom friction coefficient (λ_0) with thick lines used for width constricted channels and thin lines for depth constricted channels. It can be seen in Fig. 4(a) and (b) that an increase in either bottom drag or constriction factor of a Type I channel will cause the potential to diminish in all instances. This decrease in potential is primarily due to an enhancement in sea floor related drag and the associated decrease in channel flow rate.

Table 1
Approximate geometry values for the Cook Strait and Minas Passage. The lagoon factor for the Minas Passage was taken from Ref. [8] while the rest of the data was taken by projecting the channels onto a regular *xy*-grid. Data for the Cook Strait assumes a flat sea floor*.

Channel	Cook strait	Minas passage
L, W_0, h_0	117 km, 125 km, 100 m	9 km, 7.1 km, 71 m
Mean W, h	77 km, 100 m*	5.9 km, 59 m
W_c, h_c	0.60, 0*	0.35, 0.28
$\lambda_0 = \widehat{\alpha}LC_D/h_0$	0.13	7.00
I, B, β	1.76, 3.31, –	1.15, 1.59, 12.9
$B\lambda_0$	0.43	11.1

Bottom drag is proportional to the local velocity squared and hence a localised region of high velocity within a channel will cause the effective bottom drag coefficient of the channel as a whole to increase significantly (Fig. 4(d)). As more power is consequently lost to drag from the sea floor, less power is then available for the production of electricity.

The second contributing factor to this decrease in potential with constriction is channel transport. Transport through the channel decreases with constriction due to this aforementioned increase in bottom drag and geometric factor I. This is discussed further in Section 5 below. Results here suggest that a trade off exists in Type I channels between having a local zone of high velocity and having a high channel potential.

Fig. 4 demonstrates that depth constricted channels always have lower potentials than channels constricted in their width to the same degree. Making a channel shallower causes a greater increase in effective bottom drag than does making the channel narrower (Fig. 4). Mathematically, this is due to the $1/h$ component of the effective bottom drag term. Physically, it's due to the boundary layer associated with the sea floor extending vertically through a greater percentage of the total water column. As the channel is made shallower, a greater percentage of the channel transport is affected by this boundary layer. Although beyond the scope of this study, the presence of this boundary layer will have implications for turbines mounted on the sea floor and, in the case of shallow channels with strong tidal flows, will also affect tidal turbines floating on the surface.

The above finding suggests that, although a radical idea in principle, if a channel were to be geometrically modified to create a local zone of high velocity, it would be preferential to constrict the channel's width rather than its depth. This would create a high velocity region that may allow developers to generate power using fewer turbines. It would also, however, come at the expense of lowering the potential of the channel. Note that this study has assumed the aspect ratio of the channel is such that the channel's width is significantly greater than its depth and thus does not consider drag from the lateral walls of the channel. In deep, thin channels where the depth is of similar or greater magnitude to the channel width, this drag source will not be negligible and may in fact dominate.

4.2. Potential of type II channels

Fig. 5 shows the channel potential in Type II channels versus channel geometric factor I. Geometric factor was used instead of constriction factor here as it better demonstrated the "power peaks" that can be observed in the figure. The three coloured lines correspond to bottom drag values of $\lambda_0 = 0.1$ (purple), $\lambda_0 = 1$ (green) and $\lambda_0 = 7$ (red). Recall that for an unconstricted channel, the geometric factor $I = 1$ and as the degree of constriction is increased, the geometric factor I approaches infinity (when the channel is totally blocked off).

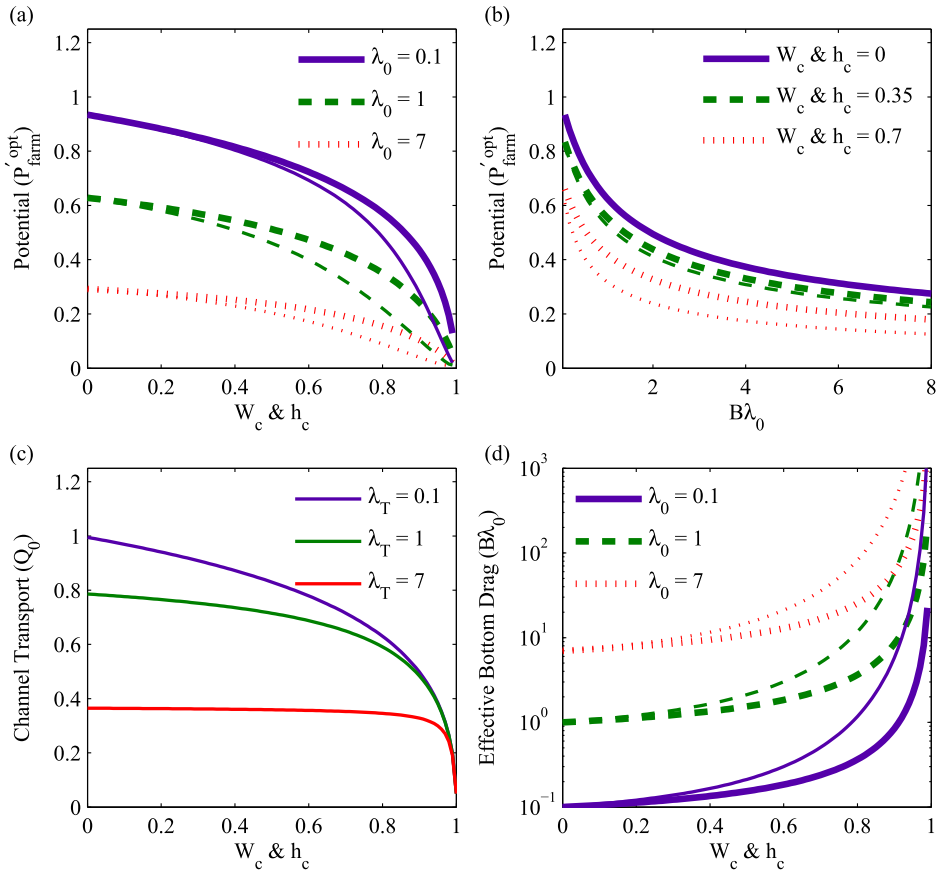

Fig. 4. Plots showing the effect of constriction and bottom drag in a Type I (open ended) channel on (a,b) channel potential, and (c) channel transport. Figure (d) shows how constriction enhances the effective bottom drag of the channel through increases in geometric factor B - note the log scale on the y axis of (d). Thick lines have been used for width constricted channels while thin lines have been used for depth constrictions. Both channel potential and transport values are relative to an unconstricted, frictionless channel.

Fig. 5 shows that the potential of a Type II channel with low bottom drag can exceed that of an unconstricted frictionless Type I channel (e.g. $\overline{P}'^{opt}_{farm} > 1$) for certain combinations of channel geometry I and lagoon geometry β. This is a consequence of feedback between channel flow rate and the free surface elevation of the terminal embayment. Certain channel dynamics and geometries give rise to enhanced head differences across the channel and consequently flow rates that are not possible in Type I channels. This feedback mechanism is believed to be the primary reason for the power peaks observed in Fig. 5.

The power peaks in Fig. 5 (c) and (d) are clearly visible. The potential of these channels increases with constriction to some peak value and then diminishes as the channel is further constricted. The peak in the upper left plot (a) lies to the left of the y axis and cannot be seen. This would suggest that channel potential could be increased by widening the channel to achieve a value of $I < 1$. Plot (b) is essentially peaking around $I = 1$ although a slight widening may possibly cause further increase in potential.

The location of these peaks on the I-axis is dependent on the lagoon geometric factor (β) of the channel and the fact that these peaks exist is an interesting result. Although an extreme proposition, the presence of these peaks suggests that Type II channels can be geometrically "tuned" to increase potential. This process could involve either constricting or widening the channel to cause an increase or decrease in the channel's I value respectively. In cases where I is below its optimal value, further constricting the channel would simultaneously increase the total power and increase the velocity in the constricted area. Such an operation is likely to be unfeasible for large scale channels such as Minas Passage, Bay of Fundy but may be feasible in smaller channels.

Analogous to observations made for Type I channels in Section 4.1 above, increasing the bottom drag coefficient of Type II channels results in a lower potential regardless of channel geometry or lagoon geometry. Again, depth constricted channels have less available power than width constricted channels of the same constriction factor.

5. Effects of constriction on flow

The degree of constriction present within a channel affects the dynamical balance of forces which has implications for the channel transport. Power is proportional to the channel's flow rate cubed and consequently the behaviour of the channel transport is pertinent to that of the potential. The approximate analytical solution described in Section 3 is a useful tool for demonstrating flow rate behaviour and is shown again below for convenience:

$$Q_0 = \frac{\sqrt{2}}{\sqrt{\sqrt{4\lambda_T^2 + (I-\beta)^4} + (I-\beta)^2}}.$$

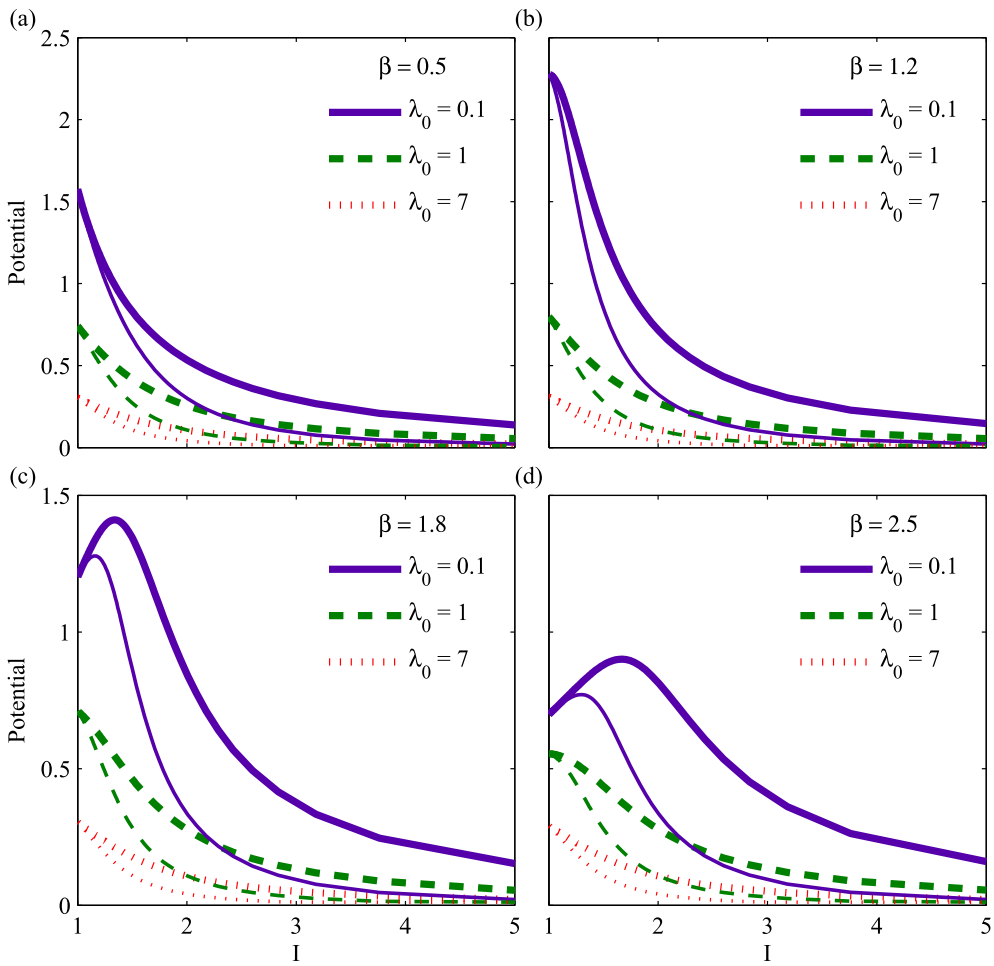

Fig. 5. Line plots of channel potential versus geometric factor I for various Type II channels of differing lagoon factors (β) and bottom friction (λ_0) configurations. Width constricted channels are denoted with thick lines while depth constricted channels are denoted with thin lines. Channel potential refers to the maximum amount of power that can be lost in a channel to turbine farm related drag and the geometric factor $I = \int_0^1 \frac{dx'}{A'}$ is determined by the amount of constriction within the channel, for an unconstricted channel $I = 1$ and this factor increases with an increase in constriction. All potential values are relative to the channel potential of a frictionless Type I channel of the same maximum dimensions.

5.1. Channels of type I

By setting $\beta = 0$, the analytical approximation above can be used for a Type I channel (Equation (16)). It is clear from (19) that both increases in drag and constriction factor (and therefore an increase in I) will cause the peak flow rate through the channel to diminish. Channel drag and channel constriction both act to retard the flow rate through a Type I channel (Fig. 4(c)). Channel drag acts to retard the flow via an increase in the λ_T term in the denominator of (19). Due to the quadratic dependence of drag on velocity, the zone of high velocity flow through the constricted passage equates to a zone of high drag:

$$Q_0 = \frac{\sqrt{2}}{\sqrt{\sqrt{4\lambda_T^2 + I^4} + I^2}}. \tag{19}$$

The presence of the constriction decreases the flow through the channel via an increase in geometric factor I. The presence of a constriction reduces the total volume water in the channel as some of the water volume is now occupied by land. As the degree of forcing in the channel is unaffected by the channel's geometry, this suggests that the flow rate should increase. However, more of the channel's forcing is now expended by accelerating the flow through the constricted passage. This latter effect dominates the volume reduction effect and while a channel constriction causes local velocity to increase, this comes at the cost of lower transport throughout the channel as a whole.

5.2. Channels of type II

As was the case for power, the relationship for Type II channels is made more complex by the feedback mechanism between channel flow rate and the free surface of the lagoon (Fig. 6 (a) - (d)). "Flow peaks" result for specific combinations of channel geometry and lagoon geometry as certain values of Q can increase the head difference at the channel ends and alter the channel forcing. Channel transport in Type II channels can exceed that of a frictionless, unconstricted Type I channel as demonstrated by peak flows greater than $Q_0 > 1$ (Fig. 6(a) - (d)). The flow rate of Type II channels is maximised when the channel's geometric factor I is equal to the lagoon geometric factor β which, in the case of zero friction, corresponds to a resonant mode in the channel-lagoon system. Although this rather elegant condition will maximise the channel's flow rate, it will not necessarily maximise the channel's potential (Fig. 5).

Applying the condition $\beta = I$ to the analytical approximation yields (20). At this condition, transport through the channel is maximised as $\lambda_T \to 0$. Increasing the amount of drag in the channel

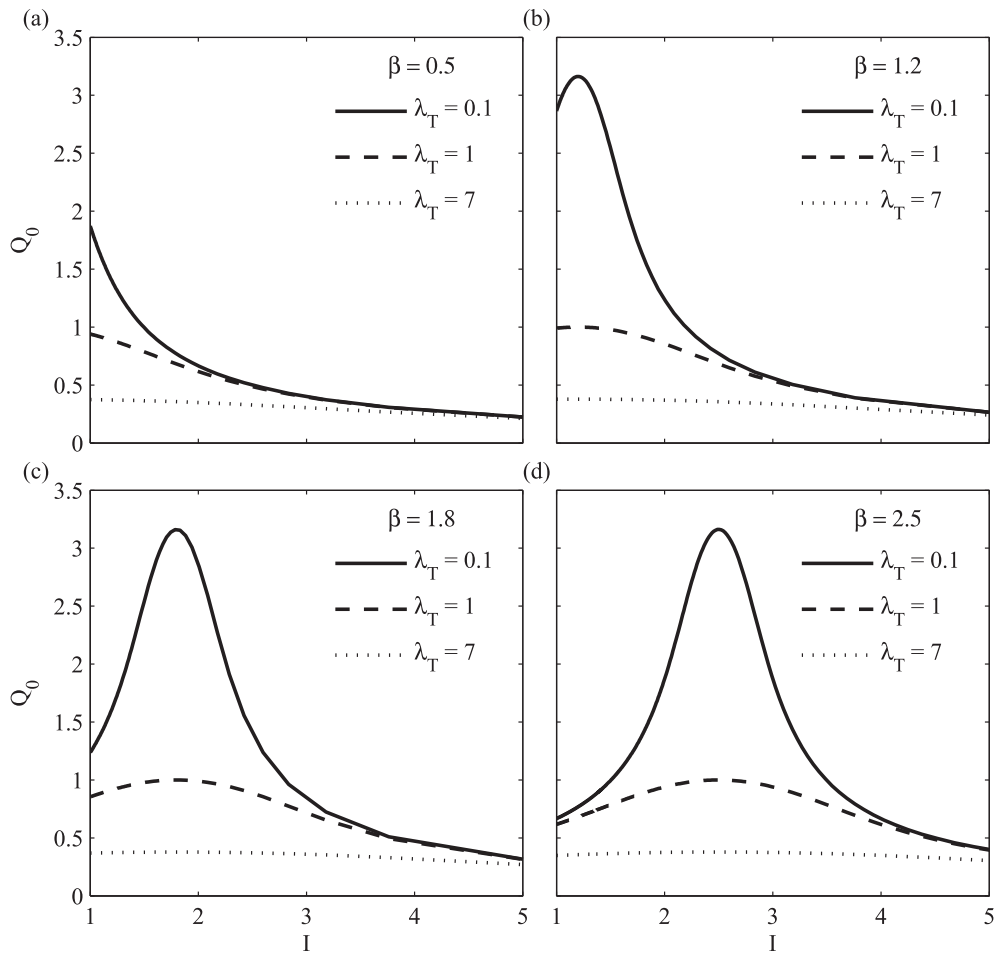

Fig. 6. Plots of the peak undisturbed flow rate Q_{0UD} versus geometric factor I for various Type II channels of differing lagoon factors (β) and total drag (λ_T) values. Note that total drag may constitute drag from both the sea floor and a turbine array. All flow rates are relative to an unconstricted, frictionless channel of the same maximum dimensions. Flow rates peak in channels where the channel geometric factor I is equal to the lagoon geometric factor β for a given value of total drag.

results in a decrease in flow rate for all geometric combinations. Any deviation from the optimum geometric condition seen in (20) causes peak flow rate to decrease:

$$Q_0 = \frac{1}{\sqrt{\lambda_T}}. \tag{20}$$

5.3. Flow reduction at potential

The negative feedback effect of adding turbines on channel flow rate is well known [2,6,8,15,16]. Equations (19) and (20) both show adding turbines (and hence adding sources of drag) to a channel will cause a decrease in peak flow rate in all instances. Fig. 7 shows the reduction in flow rate required to achieve potential in a Type I channel (a) and several Type II channels of differing lagoon geometries (b) - (f). Surprisingly, the reduction in flow rate at maximum power extraction is reasonably invariant to channel type and constriction factor. The reduction in flow rate at potential for all channels increases rapidly with small increases in effective bottom drag initially but eventually plateaus to a constant value. This trend is observed in all instances where β is less than about 3 (Fig. 7 (a) - (e)). Beyond this value of β, channels exhibit more erratic behaviour (Fig. 7 (f)) and the relationship between constriction, lagoon geometry and the balance of forces in the channel is unclear. Despite this, the flow rate reduction observed in these high β channels remains in the ballpark range of 0.6–0.7, the same range as all plots shown in Fig. 7. All the flow rate reductions shown in Fig. 7 would probably be considered unacceptable for environmental reasons [9]. Due to the obvious barriers to achieving channel potential and the relatively insignificant change in the values of flow rate reduction at high β, we have chosen not to investigate this detail any further in this work.

6. Extending the GC05 approximation

[6] provided a simple approximation for the potential of unconstricted channels which uses only the amplitude of the tidal forcing (or difference in head between the two water bodies) Δ and the undisturbed flow rate Q_{0UD} of the channel [2]. extended this approximation to an unconstricted channel connecting a bay to the open ocean:

$$\overline{P}_{drag}^{max} \simeq \gamma \rho g \Delta Q_{0UD}. \tag{21}$$

Depending on the level of bottom friction in the channel [6], found that the multiplier γ varies between 0.21 and 0.24 for a Type I channel and findings in Ref. [2] found that for a Type II channel this coefficient varies between 0.19 and 0.26 depending on bottom friction and the β parameter. Both studies find that as bottom friction increases, γ converges to near 0.22, and suggest that this

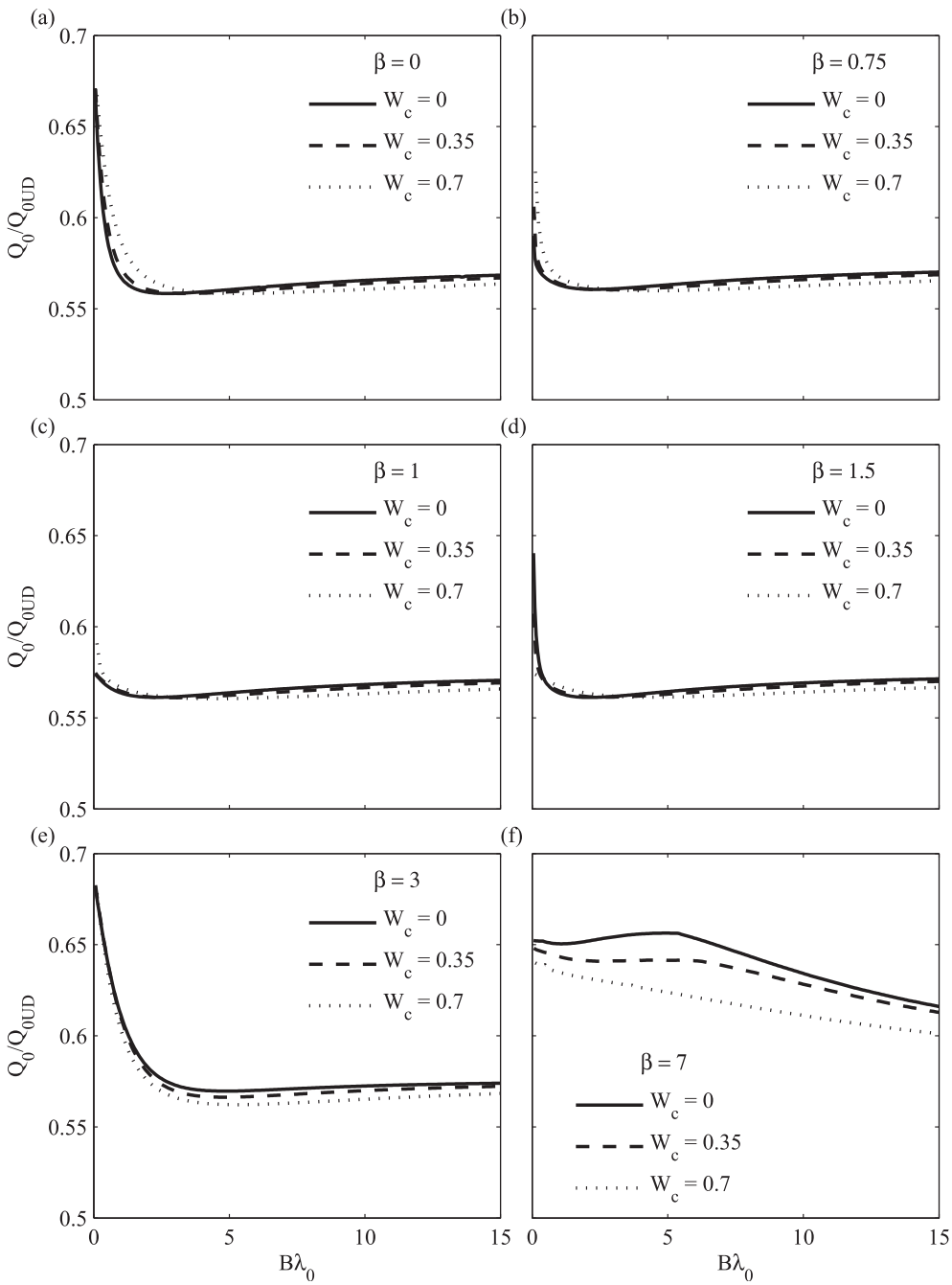

Fig. 7. Plots showing the percentage reduction in peak transport at channel potential (maximum power) for a Type I channel (a) and various Type II channels (b–f) with various constriction factors. Surprisingly the percentage reduction does not vary significantly with constriction factor and is relatively invariant to lagoon geometry provided $\beta 3$. Plots are shown only for width constrictions as the flow reduction showed little to no sensitivity to constriction type.

value may be used as a reasonable guess without any need to understand the influence of friction within the channel.

Here, this approximation was investigated for Type I and Type II channels of varying constriction factor and was found to hold with approximately the same ranges of γ given by Refs. [6,2]. The multiplier (Fig. 8) was invariant to constriction type as the channel constriction is accounted for in the calculation of B and hence the effective bottom drag ($B\lambda_0$) of the channel.

Gamma values consistently converged to a value of approximately 0.22 until β was increased beyond a value of around 7. Past this point, both the range of the multiplier and the point of convergence increased slightly.

The range of the multiplier seems to depend on how close the lagoon geometry is to the case of $\beta = 1$. Values closer to this point have a smaller range and all multiplier values lie around the convergent value of $\gamma = 0.22$ while other cases can vary by up to 0.06. As most channels lie beyond this inertial domain of bottom drag coefficients, these findings reaffirm the statement given in Refs. [6,2] that a multiplier value of 0.22 can be used with (21) as a rough approximation to the channel potential.

7. Predicting the drag coefficient at potential

At potential, the ratio of the channel bottom drag coefficient to

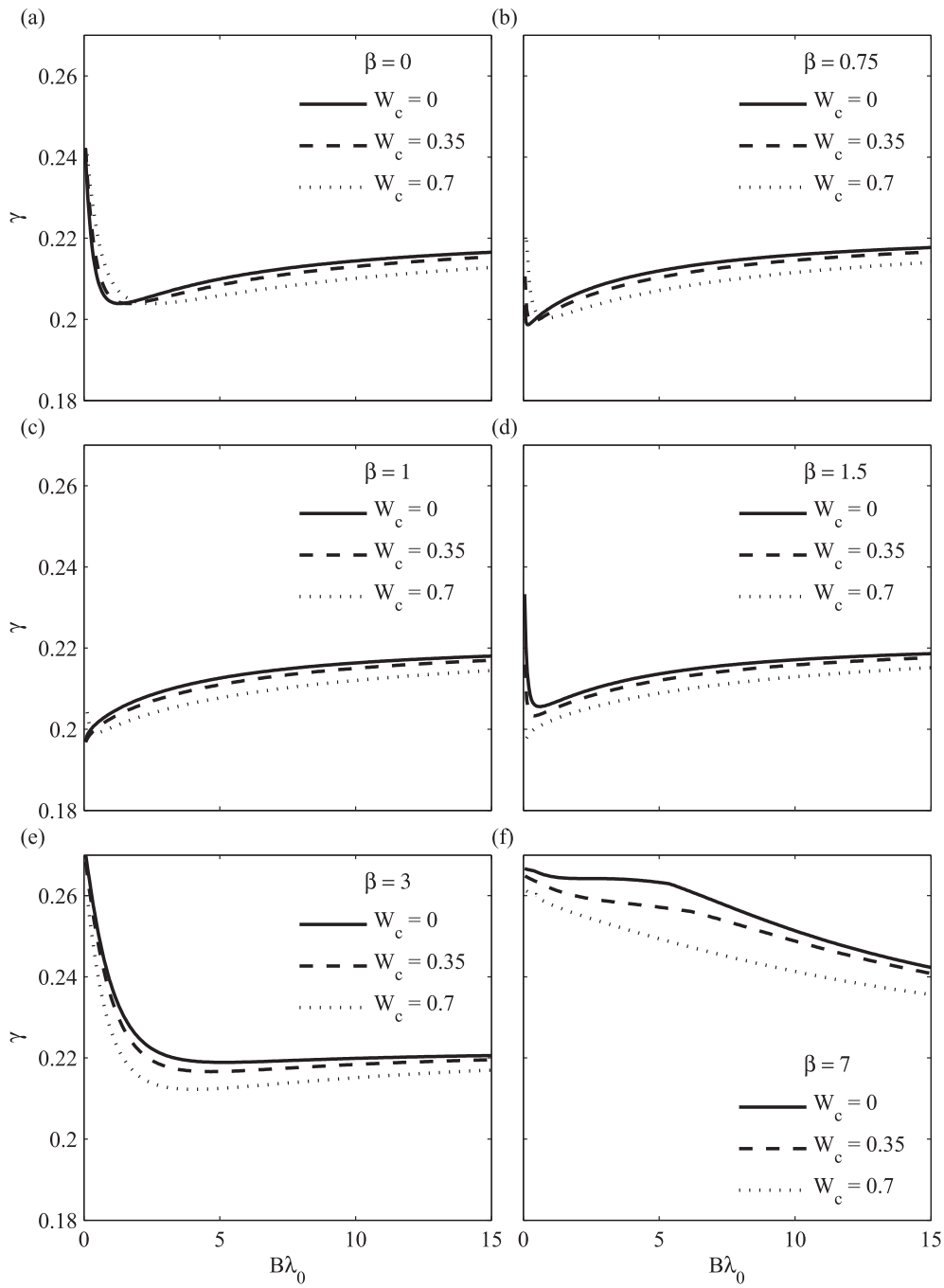

Fig. 8. Plots of the multiplier γ as introduced in Equation (21) for a variety of β values, effective bottom drag values and constriction factors. Plot (a) corresponds to a Type I channel while the remaining plots (b–f) correspond to Type II channels with different lagoon geometries. Plots are shown only for width constrictions as little to no sensitivity was observed between the multiplier γ and constriction type.

the total drag coefficient (termed $\kappa = B\lambda_0/\lambda_T$) exhibits similar behaviour to that of the multiplier mentioned above in Section 6. That is, this ratio tends to converge to a given value (0.32) as the effective bottom drag coefficient of the channel ($B\lambda_0$) increases. As was the case for the γ multiplier, the range of this value is smaller and the values of κ converge faster when the lagoon geometry factor β is close to $\beta = 1$. It can be seen in Fig. 9(f) for the Type II channel where $\beta = 7$ that the ratio does not converge in the bottom drag coefficient range shown although may do so at higher effective bottom drag. While this drag approximation is not useful for inertia dominated channels it does allow for rapid approximation of the required farm drag coefficient for shallower friction dominated channels - or at least rapid calculation of a good first guess for an optimisation routine. The total drag required to optimise a friction dominated channel can be approximated using (22) and the required drag coefficient of the turbine farm can be approximated using (23):

$$\lambda_T \approx \frac{B\lambda_0}{\kappa} \tag{22}$$

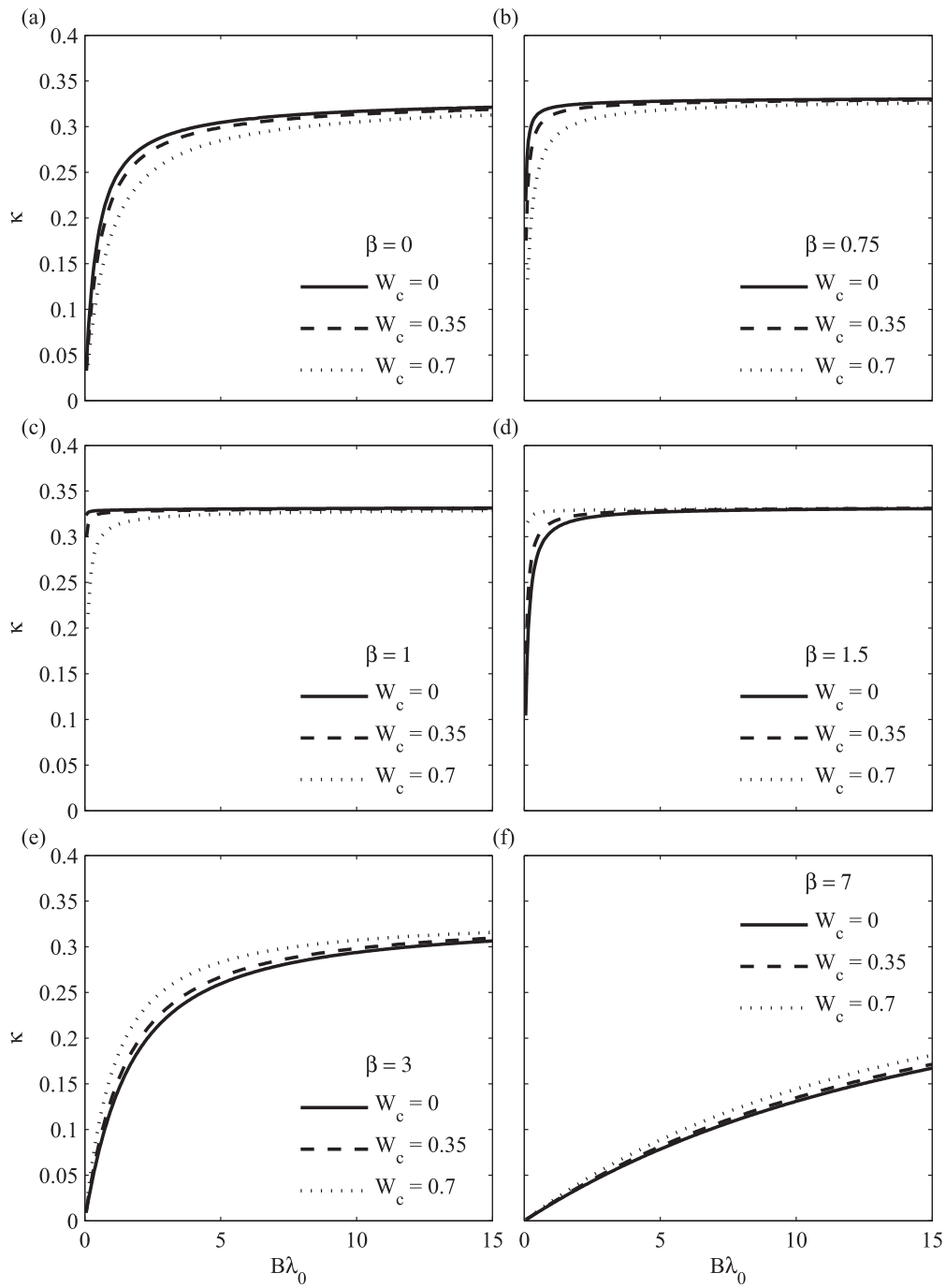

Fig. 9. Plots showing the ratio $\kappa = B\lambda_0/\lambda_T$. This value converges to a value of about 0.32 which allows for quick calculation of the farm drag coefficient required to achieve potential in shallow friction dominated channels. It can be seen in plots (b), (c) and (d) that κ also can be used for inertia dominated channels with some caution provided the β factor is close to a value of $\beta = 1$. Much like the multiplier γ, as lagoon factors deviated away from $\beta = 1$, κ was more dependent on the effective bottom drag of the channel. Plots are shown only for width constrictions as κ exhibited little or no sensitivity to constriction type.

$$\lambda_F \approx \frac{B\lambda_0 - \kappa}{\kappa} \tag{23}$$

Little distinction can be made between width constricted and depth constricted channels in the value of κ.

8. Conclusion

This paper has used a 1D shallow water model to investigate the effects of channel constriction on the maximum available power in tidal channels connecting two large water bodies (Type I) and a large water body connected to an embayment (Type II). Findings from this study indicate that increasing the degree of constriction in a Type I channel decreases the potential of the channel to produce power. This is due to an increase in the effective bottom drag of the channel and an associated decrease in the peak flow rate of the channel. As drag force is proportional to the square of the velocity, the enhanced velocities through the narrow constricted zone result in an increase in effective bottom drag coefficient of the

channel. More power is expended overcoming sea floor related drag and less is then available for electricity generation, lowering the channel's potential.

The addition of a constriction also results in a smaller flow volume through the channel. This is due to the aforementioned increase in effective bottom drag through the channel and the fact that more of the channel's forcing is now expended by accelerating the flow through the constriction. This decrease in transport through the channel further reduces channel power. Findings from this study suggest a trade off exists between having a constricted channel with a local zone of high velocity and having an unconstricted channel with a high potential.

The relationship between channel potential and constriction in Type II channels is more complicated because the free surface elevation in the terminal bay and the channel flow rate are mutually dependent. This feedback loop means that increasing the degree of constriction in the channel can cause the maximum power and flow rate of a Type II channel to either increase or decrease depending on the relative geometries of the channel and the bay. Although a somewhat radical proposition, this suggests that the channel geometry and lagoon geometry of Type II channels may be "tuned" to increase the channel's power. Furthermore, in certain instances this would have the dual effect of increasing the channel's potential as well as creating a localised zone of high velocity. The behaviour of peak flow rate in Type II channels was analogous to the behaviour of potential in these channels. The transport of Type II channels peaks when the channel geometric factor I is equal to the lagoon geometric factor β. The geometric condition that maximises channel potential is not clear.

For both types of channel examined, negative effects on potential were more profound for depth constrictions than width constrictions of equal magnitude (in terms of constriction volume). This suggests that artificially constricting a channel would preferentially be done by modifying its width rather than depth. This would create a localised zone of high velocity which may be advantageous for power extraction but would come at the cost of lowering the upper power limit of the channel as a whole.

The degree of constriction was found to have little to no effect on the flow rate reduction required to achieve channel potential in Type I channels, and, Type II channels with small β values ($\beta < 3$). For channels with low effective bottom friction coefficients ($B\lambda_0 < 3$), the necessary flow rate reduction to achieve potential increased rapidly from a value of about 33% in the near frictionless case to a value of about 43%. For higher levels of effective bottom friction ($B\lambda_0 \geq 3$), the flow rate reduction at potential for these channels exhibited little sensitivity to constriction factor, did not necessarily increase with greater effective bottom drag coefficient and did not converge to a constant value within the range tested.

Finally, the simple approximation for channel potential provided by Refs. [6,2] was found to hold for constricted channels of both Type I and Type II and a similar relationship was found to approximate the drag coefficient required by a turbine farm to achieve potential for a friction dominated channel ($B\lambda_0 > 3$). Both relationships, particularly the latter, were found to be poorer approximations for Type II channels with particularly high lagoon geometric factors ($\beta > 6$).

This study has taken, what may be the first, systematic look at the effects of channel constriction on the upper limit of tidal power production. The focus of further work will determine more realistic values of available power by building a turbine model into the pre-existing model to test various array scenarios in constricted channels. Further work is also being undertaken to expand analysis into two dimensions.

Acknowledgements

This work was supported by New Zealand Marsden Fund Grant number 12-UO-101. The authors also acknowledge Richard Karsten of Acadia University for the use of geometric data for the Minas Passage.

References

[1] Taa Adcock, The available power from tidal stream turbines in the Pentland Firth, Proc. R. Soc. A Math. Phys. Eng. Sci. 469 (2157) (2013) 1471–2946, http://dx.doi.org/10.1098/rspa.2013.0072. ISSN 1364-5021, http://rspa.royalsocietypublishing.org/content/469/2157/20130072.short.

[2] J. Blanchfield, C. Garrett, P. Wild, A. Rowe, The extractable power from a channel linking a bay to the open ocean. Proceedings of the Institution of Mechanical Engineers, Part A J. Power Energy 222 (3) (2008) 289–297, http://dx.doi.org/10.1243/09576509JPE524. ISSN 0957-6509.

[3] R. Carballo, G. Iglesias, A. Castro, Numerical model evaluation of tidal stream energy resources in the R??a de Muros (NW Spain), Renew. Energy 34 (6) (2009) 1517–1524, http://dx.doi.org/10.1016/j.renene.2008.10.028. ISSN 09601481.

[4] Scott Draper, Thomas A.A. Adcock, Alistair G.L. Borthwick, Guy T. Houlsby, Estimate of the tidal stream power resource of the Pentland Firth, Renew. Energy 63 (2014) 650–657, http://dx.doi.org/10.1016/j.renene.2013.10.015. ISSN 09601481.

[5] Matthew C. Easton, David K. Woolf, Peter a. Bowyer. The dynamics of an energetic tidal channel, the Pentland Firth, Scotland, Cont. Shelf Res. 48 (2012) 50–60, http://dx.doi.org/10.1016/j.csr.2012.08.009. ISSN 02784343.

[6] Chris Garrett, Patrick Cummins, The power potential of tidal currents in channels, Proc. R. Soc. A Math. Phys. Eng. Sci. 461 (2060) (2005) 2563–2572, http://dx.doi.org/10.1098/rspa.2005.1494. ISSN 1364-5021.

[7] Chris Garrett, Patrick Cummins, The efficiency of a turbine in a tidal channel, J. Fluid Mech. 588 (2007) 243–251, http://dx.doi.org/10.1017/S0022112007007781. ISSN 0022-1120.

[8] R.H. Karsten, J.M. McMillan, M.J. Lickley, R.D. Haynes, Assessment of tidal current energy in the Minas passage, bay of Fundy, Proc. Institution Mech. Eng. Part A J. Power Energy 222 (5) (2008) 493–507, http://dx.doi.org/10.1243/09576509JPE555. ISSN 0957-6509.

[9] Mark A. Shields, David K. Woolf, Eric P.M. Grist, Sandy A. Kerr, A.C. Jackson, Robert E. Harris, Michael C. Bell, Robert Beharie, Andrew Want, Emmanuel Osalusi, Stuart W. Gibb, Jonathan Side, Marine renewable energy: the ecological implications of altering the hydrodynamics of the marine environment, Ocean Coast. Manag. 54 (1) (2011) 2–9, http://dx.doi.org/10.1016/j.ocecoaman.2010.10.036. ISSN 09645691.

[10] Richard P. Signell, W. Rockwell Geyer, Transient eddy formation around headlands, J. Geophys. Res. 96 (1991) 2561, http://dx.doi.org/10.1029/90JC02029. ISSN 0148-0227, http://www.agu.org/pubs/crossref/1991/90JC02029.shtml.

[11] C.L. Stevens, M.J. Smith, B. Grant, C.L. Stewart, T. Divett, Tidal energy resource complexity in a large strait: the Karori Rip, Cook Strait, Cont. Shelf Res. 33 (2012) 100–109, http://dx.doi.org/10.1016/j.csr.2011.11.012. ISSN 02784343.

[12] Ross Vennell, Observations of the phase of tidal currents along a strait, J. Phys. Oceanogr. 28 (8) (1998a) 1570–1577, http://dx.doi.org/10.1175/1520-0485(1998)028<1570:OOTPOT>2.0.CO;2. ISSN 0022-3670.

[13] Ross Vennell, Oscillating barotropic currents along short channels, J. Phys. Oceanogr. 28 (8) (1998b) 1561–1569, http://dx.doi.org/10.1175/1520-0485(1998)028<1561:OBCASC>2.0.CO;2. ISSN 0022-3670.

[14] Ross Vennell, Tuning tidal turbines in-concert to maximise farm efficiency, J. Fluid Mech. 671 (2011a) 587–604, http://dx.doi.org/10.1017/S0022112010006191. ISSN 0022-1120.

[15] Ross Vennell, Estimating the power potential of tidal currents and the impact of power extraction on flow speeds, Renew. Energy 36 (12) (2011b) 3558–3565, http://dx.doi.org/10.1016/j.renene.2011.05.011. ISSN 09601481.

[16] Ross Vennell, The energetics of large tidal turbine arrays, Renew. Energy 48 (2012) 210–219, http://dx.doi.org/10.1016/j.renene.2012.04.018. ISSN 09601481.

[17] Ross Vennell, Exceeding the Betz limit with tidal turbines, Renew. Energy 55 (2013) 277–285, http://dx.doi.org/10.1016/j.renene.2012.12.016. ISSN 09601481.

[18] Ross Vennell, Simon W. Funke, Scott Draper, Craig Stevens, Tim Divett, Designing large arrays of tidal turbines: a synthesis and review, Renew. Sustain. Energy Rev. 41 (2015) 454–472, http://dx.doi.org/10.1016/j.rser.2014.08.022. ISSN 13640321.